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MCY 612

Assignment 3 AWS Inspector

April, 2025

MCY 612 Cloud Security Assignment 3

In this assignment, I used Amazon Inspector to perform security assessments on an EC2 instance.

Report

Introduction

I learned about AWS Inspector, a security evaluation tool that assists in locating weaknesses and departures from recommended practices on AWS resources. I automated the security assessment process by configuring and running Inspector on an EC2 instance using the AWS CLI. I was able to learn how to read Inspector reports to enhance instance security and obtain insights into the instance's security posture by using this to check for CVE results.

Materials

AWS Cloudshell terminal

AWS CLI

AWS Version

aws-cli/2.23.13 Python/3.12.6 Linux/6.1.127-135.201.amzn2023.x86_64 exec-env/CloudShell
exe/x86_64.amzn.2023

AWS Configure

AWS Access Key ID [*****FLZO2]

AWS Secret Access Key [*****23fZK]

Default region name [us-east-1]

Default output format [json]

Security Group sg-*****602b

VPC vpc-*****6175

Subnet ID subnet-*****6ede

Methods

Start my AWS CloudShell terminal. Create an instance in the subnet of my VPC that i created in lab 2

```
~ $ aws ec2 run-instances --image-id ami-0ee02acd56a52998e --count 1 --instance-type t3.micro --key-name carrd12-key --security-group-ids sg-0b1cfa43fd7ea02b --subnet-id subnet-09df9a6645ae0bede --tag-specifications 'ResourceType=instance,Tags=[{Key=carrd12,Value=carrd12_lab3}]'
{
  "ReservationId": "r-0084fa28dc8439d01",
  "OwnerId": "881490118823",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "50059e3d-0517-46d9-9ae1-9538d01b52d7",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-04-06T20:10:51+00:00",
            "AttachmentId": "eni-attach-059e08ab8eab07c51",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0b1cfa43fd7ea02b",
              "GroupName": "carrd12"
            }
          ]
        }
      ]
    }
  ]
}
```

```
      "GroupName": "carrd12"
    }
  ],
  "Ipv6Addresses": [],
  "MacAddress": "12:3e:28:71:d7:27",
  "NetworkInterfaceId": "eni-053677758e6cc1dd9",
  "OwnerId": "881490118823",
  "PrivateIpAddress": "10.6.8.8",
  "PrivateIpAddresses": [
    {
      "Primary": true,
      "PrivateIpAddress": "10.6.8.8"
    }
  ],
  "SourceDestCheck": true,
  "Status": "in-use",
  "SubnetId": "subnet-09df9a6645aea6ede",
  "VpcId": "vpc-0cb3e30db617f6175",
  "InterfaceType": "interface",
  "Operator": {
    "Managed": false
  }
}
],
"RootDeviceName": "/dev/sda1",
"RootDeviceType": "ebs",
"SecurityGroups": [
  {
    "GroupId": "sg-0b1cfa43fd7ea602b",
```

```
        "GroupId": "sg-0b1cfa43fd7ea602b",
        "GroupName": "carrd12"
    },
    ],
    "SourceDestCheck": true,
    "StateReason": {
        "Code": "pending",
        "Message": "pending"
    },
    "Tags": [
        {
            "Key": "carrd12",
            "Value": "carrd12_lab3"
        }
    ],
    "VirtualizationType": "hvm",
    "CpuOptions": {
        "CoreCount": 1,
        "ThreadsPerCore": 2
    },
    "CapacityReservationSpecification": {
        "CapacityReservationPreference": "open"
    },
    "MetadataOptions": {
        "State": "pending",
        "HttpTokens": "optional",
        "HttpPutResponseHopLimit": 1,
        "HttpEndpoint": "enabled",
        "HttpProtocolIpv6": "disabled",
```

```

        "HttpProtocolIpv6": "disabled",
        "InstanceMetadataTags": "disabled"
    },
    "EnclaveOptions": {
        "Enabled": false
    },
    "PrivateDnsNameOptions": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
    },
    "MaintenanceOptions": {
        "AutoRecovery": "default"
    },
    "CurrentInstanceBootMode": "legacy-bios",
    "Operator": {
        "Managed": false
    },
    "InstanceId": "i-03a2b1cf42917a599",
    "ImageId": "ami-0ee02acd56a52998e",
    "State": {
        "Code": 0,
        "Name": "pending"
    },
    "PrivateDnsName": "ip-10-6-8-8.ec2.internal",
    "PublicDnsName": "",
    "StateTransitionReason": "",
    "KeyName": "carrrd12-key",
    "AmiLaunchIndex": 0,

```

```

        "KeyName": "carrrd12-key",
        "AmiLaunchIndex": 0,
        "ProductCodes": [],
        "InstanceType": "t3.micro",
        "LaunchTime": "2025-04-06T20:10:51+00:00",
        "Placement": {
            "GroupName": "",
            "Tenancy": "default",
            "AvailabilityZone": "us-east-1c"
        },
        "Monitoring": {
            "State": "disabled"
        },
        "SubnetId": "subnet-09df9a6645aea6ede",
        "VpcId": "vpc-0cb3e30db617f6175",
        "PrivateIpAddress": "10.6.8.8"
    }
}

```

Find the public IP of my instance

```
~ $ aws ec2 describe-instances --instance-ids i-03a2b1cf42917a599 --query "Reservations[*].Instances[*].PublicIpAddress" --output text
44.204.33.170
```

Initiate a SSH connection to my EC2 Instance

```
~ $ ssh -i devenv-key.pem ubuntu@44.204.33.170
The authenticity of host '44.204.33.170 (44.204.33.170)' can't be established.
ED25519 key fingerprint is SHA256:h9+Sp7ZDzc6CDnRhYacoQ1t190dyEalPKbQo53qI4l0.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.204.33.170' (ED25519) to the list of known hosts.
Welcome to Ubuntu 16.04.7 LTS (GNU/Linux 4.4.0-1128-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

0 packages can be updated.
0 of these updates are security updates.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-6-8-8:~$ su
```

Run `sudo -s` and `su` to elevate privileges and the following code to update the distribution

```
root@ip-10-6-8-8:~# su
root@ip-10-6-8-8:/home/ubuntu#
```

`apt-get update -y`
`apt-get dist-upgrade -y`
You may see the following.

Click the tab to move to <OK>. Then click Enter.

`apt-get upgrade -y`
`apt autoremove -y`

```
root@ip-10-6-8-8:/home/ubuntu# apt-get update -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates InRelease [106 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-backports InRelease [106 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial/universe amd64 Packages [7,532 kB]
Get:5 http://security.ubuntu.com/ubuntu xenial-security InRelease [106 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial/universe Translation-en [4,354 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial/multiverse amd64 Packages [144 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial/multiverse Translation-en [106 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 Packages [1,269 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main Translation-en [303 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/universe amd64 Packages [1,171 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/universe Translation-en [334 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/multiverse amd64 Packages [21.6 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/multiverse Translation-en [8,440 B]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-backports/main amd64 Packages [10.2 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-backports/main Translation-en [4,456 B]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-backports/universe amd64 Packages [11.5 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-backports/universe Translation-en [4,476 B]
Get:19 http://security.ubuntu.com/ubuntu xenial-security/main amd64 Packages [913 kB]
Get:20 http://security.ubuntu.com/ubuntu xenial-security/main Translation-en [211 kB]
Get:21 http://security.ubuntu.com/ubuntu xenial-security/universe amd64 Packages [740 kB]
Get:22 http://security.ubuntu.com/ubuntu xenial-security/universe Translation-en [203 kB]
Get:23 http://security.ubuntu.com/ubuntu xenial-security/multiverse amd64 Packages [7,864 B]
Get:24 http://security.ubuntu.com/ubuntu xenial-security/multiverse Translation-en [2,672 B]
Fetched 17.7 MB in 3s (5,107 kB/s)
Reading package lists... Done
root@ip-10-6-8-8:/home/ubuntu#
```

```
root@ip-10-6-8-8:/home/ubuntu# apt-get dist-upgrade -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  distro-info libjson-glib-1.0-0 libjson-glib-1.0-common python3-distro-info ubuntu-advantage-desktop-daemon ubuntu-pro-client ubuntu-pro-client-l10n
The following packages will be upgraded:
  apparmor apt apt-transport-https apt-utils base-files bind9-host distro-info-data dnstools grub-common grub-pc grub-pc-bin grub2-common libapparmor-perl libapparmor1
  libapt-inst2.0 libapt-pkg5.0 libbind9-140 libc-bin libc6 libdns-export162 libdns162 libgnutls-openssl27 libgnutls30 libisc-export160 libisc160 libisc940 libiscfg140 liblwres141
  libssl1.0.0 locales motd-news-config multiarch-support openssl python-apt-common python3-apt python3-software-properties python3-update-manager software-properties-common
  sosreport ubuntu-advantage-tools update-manager-core update-notifier-common
42 upgraded, 7 newly installed, 0 to remove and 0 not upgraded.
Need to get 17.6 MB of archives.
After this operation, 2,583 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 motd-news-config all 9.4ubuntu4.14 [4,480 B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 base-files amd64 9.4ubuntu4.14 [60.9 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 libc-bin amd64 2.23-0ubuntu11.3 [2,590 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 locales all 2.23-0ubuntu11.3 [3,197 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 libc-bin amd64 2.23-0ubuntu11.3 [629 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 libapt-pkg5.0 amd64 1.2.35 [715 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 libapt-inst2.0 amd64 1.2.35 [54.8 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 apt amd64 1.2.35 [1,107 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 apt-utils amd64 1.2.35 [196 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 grub-pc amd64 2.02-beta2-36ubuntu3.32 [197 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 grub-pc-bin amd64 2.02-beta2-36ubuntu3.32 [892 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 grub2-common amd64 2.02-beta2-36ubuntu3.32 [513 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 grub-common amd64 2.02-beta2-36ubuntu3.32 [1,789 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu xenial-updates/main amd64 python-apt-common all 1.1.0-beta1ubuntu0.16.04.12 [16.7 kB]
```

```
root@ip-10-6-8-8:/home/ubuntu# apt-get upgrade -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
root@ip-10-6-8-8:/home/ubuntu# apt autoremove -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Install the AWS Inspector Agent in the EC2 Instance

```
root@ip-10-6-8-8:/home/ubuntu# wget https://inspector-agent.amazonaws.com/linux/latest/install
--2025-04-06 20:26:18-- https://inspector-agent.amazonaws.com/linux/latest/install
Resolving inspector-agent.amazonaws.com (inspector-agent.amazonaws.com)... 3.167.85.211, 2600:9000:27cb:8e00:4:6195:f555:501, 2600:9000:27cb:f800:4:6195:f555:501, ...
Connecting to inspector-agent.amazonaws.com (inspector-agent.amazonaws.com)|3.167.85.211|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 31083 (30K) [binary/octet-stream]
Saving to: 'install'

install                                100%[=====] 30.35K  --.-KB/s  in 0s

2025-04-06 20:26:18 (141 MB/s) - 'install' saved [31083/31083]
```

```

root@ip-10-6-8-8:/home/ubuntu# bash install
Ubuntu.16.04
ubuntu.16.04.16
Distribution of the machine is Ubuntu.
Distribution type of the machine is ubuntu.
Revision of the distro is 16.04.
Kernel version of the machine is 4.4.0-1128-aws.
hostname: Name or service not known
is an EC2 instance reporting region as us-east-1.
dpkg-query: no packages found matching awsagent
Check for existing AwsAgent completed with error code:0
Existing version of agent installed is 0.0.0.0.
dpkg-query: no packages found matching awsagentkernelmodule-4.4.0-1128-aws
cannot find installed KM VERSION..., assumed "0.0.0.0"
get-installed_km_version: EXISTING_KM_VERSION:0.0.0.0 exit status:0
Existing version of kernel module installed is 0.0.0.0.
PUBKEY FILE: /tmp/awsagent.hyphtray/inspector.gpg
Validated AWS_AGENT_INVENTORY signature with: Primary key fingerprint: DDA0 D4C5 10AE 3C20 6F46 6DC0 2474 0960 5836 0418
HTTP/1.1 200 OK
x-amz-id-2: Cxt/YNeLwVtZPY7SSUzhjGSYQv3WzOCGvmJTDADpn70i2CMcDHFwDwrtuvN0c1IwIgwG6GZ4jrxJmnyWmsw==
x-amz-request-id: FZ7FANB6R4RQ3BY6
Date: Sun, 06 Apr 2025 20:26:28 GMT
Last-Modified: Wed, 01 Feb 2023 21:31:09 GMT
ETag: "c9e3db376ee55a33c58a6a2e419b823"
x-amz-server-side-encryption: AES256
Accept-Ranges: bytes
Content-Type: binary/octet-stream
Content-Length: 792
Server: AmazonS3

```

Check the instpector agent

```

root@ip-10-6-8-8:/home/ubuntu# /etc/init.d/awsagent status
Configuration file path : /opt/aws/awsagent/etc/agent.cfg
Configuration status :
Proxy In Use: false
Agent version : 1.1.1921.0
System release : "Ubuntu 16.04.7 LTS" - 4.4.0-1128-aws
Daemon : true
Max queue size : 157286400
Log File :
Msgs log max size : 33554432
Groups : false
Users : false
CodeModules : false
InstanceMetaData : false
PackageInfo : false
NetworkInterfaces : false
Terminals : false
ConfigurationInfo : false
Permissions : false
KernelModules : false
Oval : false
OpenPorts : false
System performance : false
System performance update frequency : 0
System performance send frequency : 0
Process performance : false
Process performance update frequency : 0
Process performance send frequency : 0
Kernel module search path : /opt/aws/awsagent/kmods/amznmon64.ko

```

Create a resource group

“resourceGroupArn”: “arn:aws:inspector:us-east-1:881490118823:resourcegroup/0-Dg2puyir”

```

~ $ aws inspector create-resource-group --resource-group-tags key=carrd12,value=carrd12_lab3
{
  "resourceGroupArn": "arn:aws:inspector:us-east-1:881490118823:resourcegroup/0-Dg2puyir"
}
~ $

```


Creates a new assessment target

"assessmentTargetArn": "arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s"

```
~ $ aws inspector create-assessment-target --assessment-target-name carrd12-lab3 --resource-group-arn arn:aws:inspector:us-east-1:881490118823:resourcegroup/0-Dg2puyir
{
  "assessmentTargetArn": "arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s"
}
~ $
```

List all available Amazon Inspector rules packages.

```
~ $ aws inspector list-rules-packages
{
  "rulesPackageArns": [
    "arn:aws:inspector:us-east-1:316112463485:rulespackage/0-PmNV0Tcd",
    "arn:aws:inspector:us-east-1:316112463485:rulespackage/0-R01qwB5Q",
    "arn:aws:inspector:us-east-1:316112463485:rulespackage/0-gEjTy7T7",
    "arn:aws:inspector:us-east-1:316112463485:rulespackage/0-rExsr2X8"
  ]
}
~ $
```

create an assessment template

"assessmentTemplateArn": "arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6"

arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS/finding/0-0K
OzMk5A

```
~ $ aws inspector create-assessment-template --assessment-target-arn arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s --assessment-template-name carrd12-lab3 --duration-in-seconds 180 --rules-package-arns arn:aws:inspector:us-east-1:316112463485:rulespackage/0-gEjTy7T7 --user-attributes-for-findings key=carrd12,value=carrd12-assessment
{
  "assessmentTemplateArn": "arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6"
}
~ $
```

Start the assessment run

"assessmentRunArn": "assessmentRunArn":

"arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS"

```
~ $ aws inspector start-assessment-run --assessment-run-name InitialInspectorTrial --assessment-template-arn arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6
{
  "assessmentRunArn": "arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS"
}
~ $
```

Produce an assessment report

Download the assessment PDF report

```
~ $ aws inspector get-assessment-report --assessment-run-arn arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS --report-file-format PDF --report-type FULL
{
  "status": "COMPLETED",
  "url": "https://inspector-temp-reports-prod-us-east-1.s3.amazonaws.com/arn%3Aaws%3Ainspector%3Aus-east-1%3A881490118823%3Atarget/0-BcbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS-full-report.pdf?response-content-type=application%2Fpdf&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20250406T205825Z&X-Amz-SignedHeaders=host&X-Amz-Expires=900&X-Amz-Credential=AKIAUTG0FMZ6STOELRNNK2F20250406%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Signature=c4bd36bd05433f6f11346f0c37a9c0d0d425bc09788287a9fe65924f21156999"
}
~ $
```

Describe the first finding

"title": "Instance i-03a2b1cf42917a599 is vulnerable to CVE-2024-26677",

"severity": "Medium",

"recommendation": "Use your Operating System's update feature to update package

linux-image-4.4.0-1128-aws-0:4.4.0-1128.142, linux-image-aws-0:4.4.0-1128.133-0. For more information see

https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-26677",

```

~ $ aws inspector describe-findings --finding-arns arn:aws:inspector:us-east-1:881490118823:target/0-8cbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS/finding/0-0K0zMk5A
{
  "findings": [
    {
      "arn": "arn:aws:inspector:us-east-1:881490118823:target/0-8cbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS/finding/0-0K0zMk5A",
      "schemaVersion": 1,
      "service": "Inspector",
      "serviceAttributes": {
        "schemaVersion": 1,
        "assessmentRunArn": "arn:aws:inspector:us-east-1:881490118823:target/0-8cbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS",
        "rulesPackageArn": "arn:aws:inspector:us-east-1:316112463485:rulespackage/0-gEjTy7T7"
      },
      "assetType": "ec2-instance",
      "assetAttributes": {
        "schemaVersion": 1,
        "agentId": "i-03a2b1cf42917a599",
        "amiId": "ami-0ee0acd56a52998e",
        "hostname": "ip-10-6-8-8",
        "ipv4Addresses": [],
        "tags": [
          {
            "key": "carrd12",
            "value": "carrd12_lab3"
          }
        ],
        "networkInterfaces": [
          {
            "networkInterfaceId": "eni-053677758e6cc1dd9",

```

```

            "networkInterfaceId": "eni-053677758e6cc1dd9",
            "subnetId": "subnet-09df9a6645aea6ede",
            "vpcId": "vpc-0cb3e38db617f6175",
            "privateIpAddress": "10.6.8.8",
            "privateIpAddresses": [
              {
                "privateIpAddress": "10.6.8.8"
              }
            ],
            "publicIp": "44.204.33.170",
            "ipv6Addresses": [],
            "securityGroups": [
              {
                "groupName": "carrd12",
                "groupId": "sg-0b1cf43fd7ea682b"
              }
            ]
          }
        ],
        "id": "CVE-2024-26677",
        "title": "Instance i-03a2b1cf42917a599 is vulnerable to CVE-2024-26677",
        "description": "In the Linux kernel, the following vulnerability has been resolved:\n\nvxrpc: Fix delayed ACKs to not set the reference serial number\n\nFix the constructi\non of delayed ACKs to not set the reference serial number\n\nas they can't be used as an RTT reference.",
        "recommendation": "Use your Operating System's update feature to update package linux-image-4.4.0-1128-aws-0:4.4.0-1128.142, linux-image-aws-0:4.4.0.1128.133-0. For more i\nformation see [https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-26677](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-26677)",
        "severity": "Medium",
        "numericSeverity": 6.0,

```

```

      "numericSeverity": 6.0,
      "confidence": 10,
      "indicatorOfCompromise": false,
      "attributes": [
        {
          "key": "CVSS3_SCORE",
          "value": "5.5"
        },
        {
          "key": "package_name",
          "value": "linux-image-4.4.0-1128-aws-0:4.4.0-1128.142, linux-image-aws-0:4.4.0.1128.133-0"
        },
        {
          "key": "CVSS3_VECTOR",
          "value": "AV:L/AC:L/PR:L/UI:N/S:U/C:N/I:N/A:H"
        },
        {
          "key": "CVE_ID",
          "value": "CVE-2024-26677"
        },
        {
          "key": "INSTANCE_ID",
          "value": "i-03a2b1cf42917a599"
        }
      ],
      "userAttributes": [
        {
          "key": "carrd12",

```

```

    "key": "package_name",
    "value": "linux-image-4.4.0-1128-aws-0:4.4.0-1128.142, linux-image-aws-0:4.4.0.1128.133-0"
  },
  {
    "key": "CVSS3_VECTOR",
    "value": "AV:L/AC:L/PR:L/UI:N/S:U/C:N/I:N/A:H"
  },
  {
    "key": "CVE_ID",
    "value": "CVE-2024-26677"
  },
  {
    "key": "INSTANCE_ID",
    "value": "i-03a2b1cf42917a599"
  }
],
"userAttributes": [
  {
    "key": "carrd12",
    "value": "carrd12-assessment"
  }
],
"createdAt": "2025-04-06T20:49:51.119000+00:00",
"updatedAt": "2025-04-06T20:49:51.119000+00:00"
}
],
"failedItems": {}
}
}
(END)

```

Ssh into my EC2 instance

```

~ $ ssh -i devenv-key.pem ubuntu@44.204.33.170
Welcome to Ubuntu 16.04.7 LTS (GNU/Linux 4.4.0-1128-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

193 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 16.04 at
https://ubuntu.com/16-04

New release '18.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

*** System restart required ***
Last login: Sun Apr  6 20:21:54 2025 from 3.87.244.153
ubuntu@ip-10-6-8-8:~$

```

edit /etc/ssh/sshd_config to set PermitRootLogin to no

```

root@ip-10-6-8-8:/home/ubuntu# vi /etc/ssh/sshd_config
root@ip-10-6-8-8:/home/ubuntu#

```

```
# To enable empty passwords, change to yes (NOT RECOMMENDED)
Port 22
# Use these options to restrict which interfaces/protocols sshd will bind to
#ListenAddress ::
#ListenAddress 0.0.0.0
Protocol 2
# HostKeys for protocol version 2
HostKey /etc/ssh/ssh_host_rsa_key
HostKey /etc/ssh/ssh_host_dsa_key
HostKey /etc/ssh/ssh_host_ecdsa_key
HostKey /etc/ssh/ssh_host_ed25519_key
#Privilege Separation is turned on for security
UsePrivilegeSeparation yes

# Lifetime and size of ephemeral version 1 server key
KeyRegenerationInterval 3600
ServerKeyBits 1024

# Logging
SyslogFacility AUTH
LogLevel INFO

# Authentication:
LoginGraceTime 120
PermitRootLogin no
StrictModes yes

RSAAuthentication yes
-- INSERT --
```

Save edit using esc :wq

restart the SSH daemon

```
root@ip-10-6-8-8:/home/ubuntu# systemctl restart sshd
root@ip-10-6-8-8:/home/ubuntu#
```

exit the ssh connection to your EC2 instance

```
root@ip-10-6-8-8:/home/ubuntu# exit
exit
ubuntu@ip-10-6-8-8:~$ exit
logout
Connection to 44.204.33.170 closed.
~ $
```

Testing

Generate another assessment PDF report

Creates a new assessment target

```
~ $ aws inspector create-assessment-target --assessment-target-name your_username-lab3 --resource-group-arn arn:aws:inspector:us-east-1:881490118823:resourcegroup/0-Dg2puyir
{
  "assessmentTargetArn": "arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf"
}
```

"assessmentTargetArn": "arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf"

create an assessment template

```
~ $ aws inspector create-assessment-template --assessment-target-arn arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf --assessment-template-name carrd12-lab3 --duration-in-seconds 180 --rules-package-arns arn:aws:inspector:us-east-1:316112463485:rulespackage/0-gEjTy7T7 --user-attributes-for-findings key=carrd12,value=carrd12-assessment
{
  "assessmentTemplateArn": "arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp"
}
```

assessmentTemplateArn": "arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp"

Start the assessment run

```
~ $ aws inspector start-assessment-run --assessment-run-name InitialInspectorTrial --assessment-template-arn arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp
{
  "assessmentRunArn": "arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp/run/0-AdTUwMt8"
}
```

"assessmentRunArn": "assessmentRunArn":

"arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp/run/0-AdTUwMt8"

Produce an assessment report

```
~ $ aws inspector get-assessment-report --assessment-run-arn arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp/run/0-AdTUwMt8 --report-file-format PDF --report-type FULL
{
  "status": "COMPLETED",
  "url": "https://inspector-temp-reports-prod-us-east-1.s3.amazonaws.com/arn%3Aaws%3Ainspector%3Aus-east-1%3A881490118823%3Atarget/0-Jp115pdf/template/0-rZo9vJXp/run/0-AdTUwMt8-full-report.pdf?response-content-type=application%2Fpdf&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20250406T221932Z&X-Amz-SignedHeaders=host&X-Amz-Expires=900&X-Amz-Credential=AKIAUTGOFMZ65TOELRNW%2F20250406%2Fus-east-1%2F%3A%2Faws4_request&X-Amz-Signature=9340f1e364fa377b5c9ce1cc608d1a08e5ae50371a3fc844ee46cfdaaf1154a"
}
```



Amazon Inspector - Assessment Report

Full Report

Report generated on 2025-04-06 at 22:14:57 UTC

Assessment Template: carrd12-lab3

Assessment Run start: 2025-04-06 at 22:06:06 UTC
Assessment Run end: 2025-04-06 at 22:11:36 UTC



Amazon Inspector - Assessment Report

Full Report

Report generated on 2025-04-06 at 20:53:29 UTC

Assessment Template: carrd12-lab3

Assessment Run start: 2025-04-06 at 20:45:02 UTC
Assessment Run end: 2025-04-06 at 20:50:23 UTC

After testing my assessment target including my instance against the rule package listed i was unable to find the “Disable root login over SSH” finding.

Clean up

Terminate my instance.

```

~ $ aws ec2 terminate-instances --instance-ids i-03a2b1cf42917a599
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-03a2b1cf42917a599",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
~ $ █

```

Delete the assessment runs

```

~ $ aws inspector delete-assessment-run --assessment-run-arn arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp/run/0-AdTUwMt8
~ $ aws inspector delete-assessment-run --assessment-run-arn arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6/run/0-F8yc15rS

```

Delete the assessment templates

```

~ $ aws inspector delete-assessment-template --assessment-template-arn arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf/template/0-rZo9vJXp
~ $ aws inspector delete-assessment-template --assessment-template-arn arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s/template/0-c6w1FeJ6

```

Deletes the assessment target

```

~ $ aws inspector delete-assessment-target --assessment-target-arn arn:aws:inspector:us-east-1:881490118823:target/0-BcbTLM9s
~ $ aws inspector delete-assessment-target --assessment-target-arn arn:aws:inspector:us-east-1:881490118823:target/0-Jp115pdf
~ $ █

```

Conclusion

To investigate the security evaluation features of AWS Inspector I set up an EC2 instance. I discovered a vulnerability on the instance after installing and configuring AWS Inspector. I then used Inspector to rerun the scan after applying the required update to resolve the problem. I was able to verify that the vulnerability had been fixed and gain a better understanding of how Inspector helps in monitoring and enhancing the security posture of EC2 instances by contrasting the before and after data.

References

Security best practices for Amazon Inspector Classic - Amazon Inspector classic. (n.d.-c).
https://docs.aws.amazon.com/inspector/v1/userguide/inspector_security-best-practices.html